WORKED SOLUTION

STEP I 1987, Question 16

Probability and Statistics • Bayes and Normal Approximation

STEP I 1987, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q16

The key idea

Part (i) is Bayes' theorem in disguise. Voting the same way twice is *evidence* about which party the member belongs to, because a Preservative always does it and a Progressive only sometimes does. So we first update the chance that the member is a Preservative, and then predict the third vote. In part (ii) the Preservatives' votes are fixed numbers, so all the randomness comes from the 40 Progressives. Their number of votes in favour is binomial, and that is what we approximate by a normal distribution.

Solution

Part (i)

Let R be the event that the chosen member is a Preservative, so $P(R) = \frac{60}{100} = \frac{3}{5}$ and $P(R') = \frac{2}{5}$. Let S_2 be the event that the member voted the same way on the first two occasions, and S_3 the event that the third vote also matches.

A Preservative never changes their mind, so $P(S_2 \mid R) = 1$. A Progressive votes at random, each vote independently for or against with probability $\frac{1}{2}$. Whatever the first vote was, the second matches it with probability $\frac{1}{2}$, so $P(S_2 \mid R') = \frac{1}{2}$. Hence

$$P(S_2) = \tfrac{3}{5} \cdot 1 + \tfrac{2}{5} \cdot \tfrac{1}{2} = \tfrac{4}{5}, \qquad P(R \mid S_2) = \frac{P(R)\,P(S_2 \mid R)}{P(S_2)} = \frac{3/5}{4/5} = \frac{3}{4}.$$

So, given what we have seen, the member is a Preservative with probability $\frac{3}{4}$ and a Progressive with probability $\frac{1}{4}$.

Ask yourself Why can't we just say $P(R \mid S_2) = \frac{3}{5}$, since the member was chosen at random?

Because the information S_2 is more likely for a Preservative than for a Progressive, it shifts the odds. Before, the odds were $3 : 2$ in favour of a Preservative. After seeing S_2 they are $3 \cdot 1 : 2 \cdot \frac{1}{2} = 3 : 1$.

Now, given the member's party, the third vote does not depend on the first two votes beyond what they tell us about the party. A Preservative repeats the vote for certain, and a Progressive matches it with probability $\frac{1}{2}$. So

$$P(S_3 \mid S_2) = \tfrac{3}{4} \cdot 1 + \tfrac{1}{4} \cdot \tfrac{1}{2} = \frac{7}{8}.$$

(Equivalently, $P(S_3 \mid S_2) = \dfrac{P(S_2 \cap S_3)}{P(S_2)} = \dfrac{\frac{3}{5} + \frac{2}{5} \cdot \frac{1}{4}}{\frac{4}{5}} = \dfrac{7/10}{4/5} = \dfrac{7}{8}$.)

Part (ii)

Consider a vote on an item of Preservative policy, and assume all 100 members vote. A proportion α of the 60 Preservatives, that is 60α members, always vote against, and the remaining $60(1-\alpha)$ always vote for. Let X be the number of Progressives voting for. Each Progressive votes for with probability $\frac{1}{2}$, independently, so

$$X \sim \mathrm{B}\left(40, \tfrac{1}{2}\right), \qquad E(X) = 20, \qquad \mathrm{Var}(X) = 40 \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = 10.$$

A simple majority of 100 votes means at least 51 in favour:

$$60(1-\alpha) + X \geq 51 \iff X \geq 60\alpha - 9.$$

Since $n = 40$ is fairly large and $p = \frac{1}{2}$ makes the binomial symmetric, we approximate X by $Y \sim \mathrm{N}(20, 10)$. Here $60\alpha - 9$ is a whole number (a number of members), so with a continuity correction

$$P(\text{majority}) = P(X \geq 60\alpha - 9) \approx P(Y > 60\alpha - 9.5) = P\left(Z > \frac{60\alpha - 29.5}{\sqrt{10}}\right),$$

where $Z \sim \mathrm{N}(0,1)$.

This probability decreases as α increases, so an election is called exactly when α is large enough to push it below 0.8. From tables, $P(Z > -0.8416) = 0.8$, so we need

$$\begin{aligned} \frac{60\alpha - 29.5}{\sqrt{10}} > -0.8416 &\iff 60\alpha > 29.5 - 0.8416\sqrt{10} = 26.84 \\ &\iff \alpha > 0.447. \end{aligned}$$

Since 60α must be a whole number of members, the smallest possible value is $60\alpha = 27$. So the least value of α that results in an election is

$$\alpha = \frac{27}{60} = 0.45.$$

To see that this is right: with 26 rebels the approximate chance of a majority is $P(Z > -1.107) \approx 0.866$, and with 27 rebels it is $P(Z > -0.791) \approx 0.785 < 0.8$.

Checks

- In (i), $\frac{7}{8}$ lies between $\frac{1}{2}$ (a known Progressive) and 1 (a known Preservative), as it must. A simulation of 400,000 random members gave 0.875.
- In (ii), the exact binomial probabilities are $P(X \geq 17) = 0.8659$ (26 rebels) and $P(X \geq 18) = 0.7852$ (27 rebels). These agree with the normal values to three decimal places, and confirm that the crossover is between 26 and 27 rebels, i.e. at $\alpha = 0.45$.

Common slips

- Answering $\frac{3}{5} \cdot 1 + \frac{2}{5} \cdot \frac{1}{2} = \frac{4}{5}$ in (i). This ignores the information that the member has already voted the same way twice.
- Treating a simple majority as "at least 50". A tie of 50–50 is not a majority, so 51 votes are needed.
- Leaving out the continuity correction. Without it the threshold becomes $\alpha > 0.456$, which leads to 28 rebels instead of 27, and that is wrong according to the exact binomial calculation.

Beyond STEP

Part (i) is a small example of *Bayesian updating*: data change our belief about a hidden "type", and predictions are then averages over the updated beliefs (the posterior predictive distribution). The normal approximation in part (ii) is a case of the central limit theorem, and the near-perfect agreement here is no accident, since for $p = \frac{1}{2}$ the binomial is symmetric. Both ideas are developed carefully in Blitzstein and Hwang, *Introduction to Probability*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.